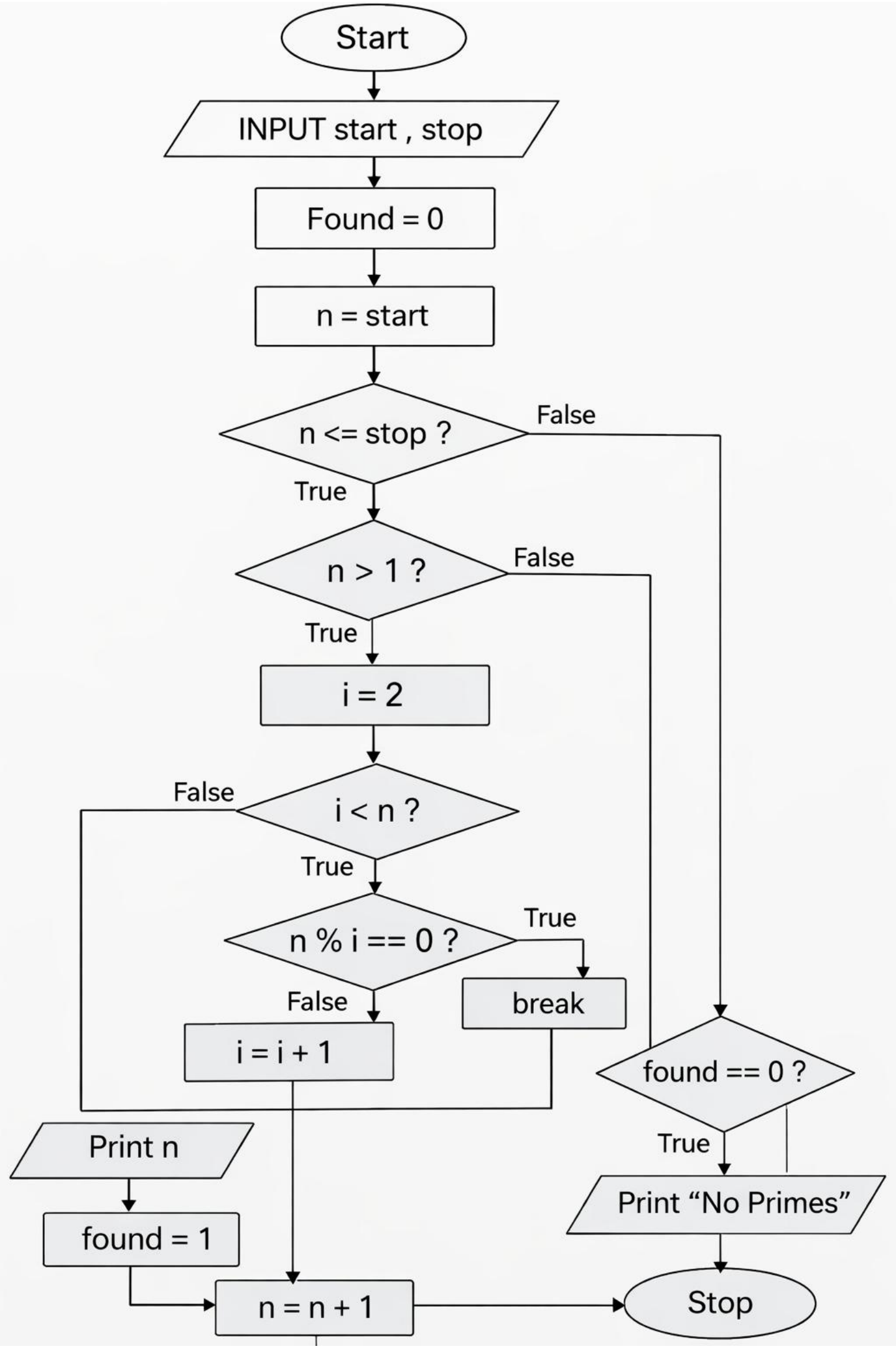
8.1.2 Prime Numbers in a Given Range

Algorithm :

Step 1: Start
Step 2: Input start and stop
Step 3: Set found = 0
Step 4: Repeat for n from start to stop
    Step 4.1: If n > 1 then
        Step 4.1.1: Repeat for i from 2 to n-1
            If n % i == 0 then
                n is not prime
                break the loop
        Step 4.1.2: If loop completes without break then
            n is prime
            Print n
            Set found = 1
Step 5: If found == 0 then
    Print "No primes"
Step 6: Stop

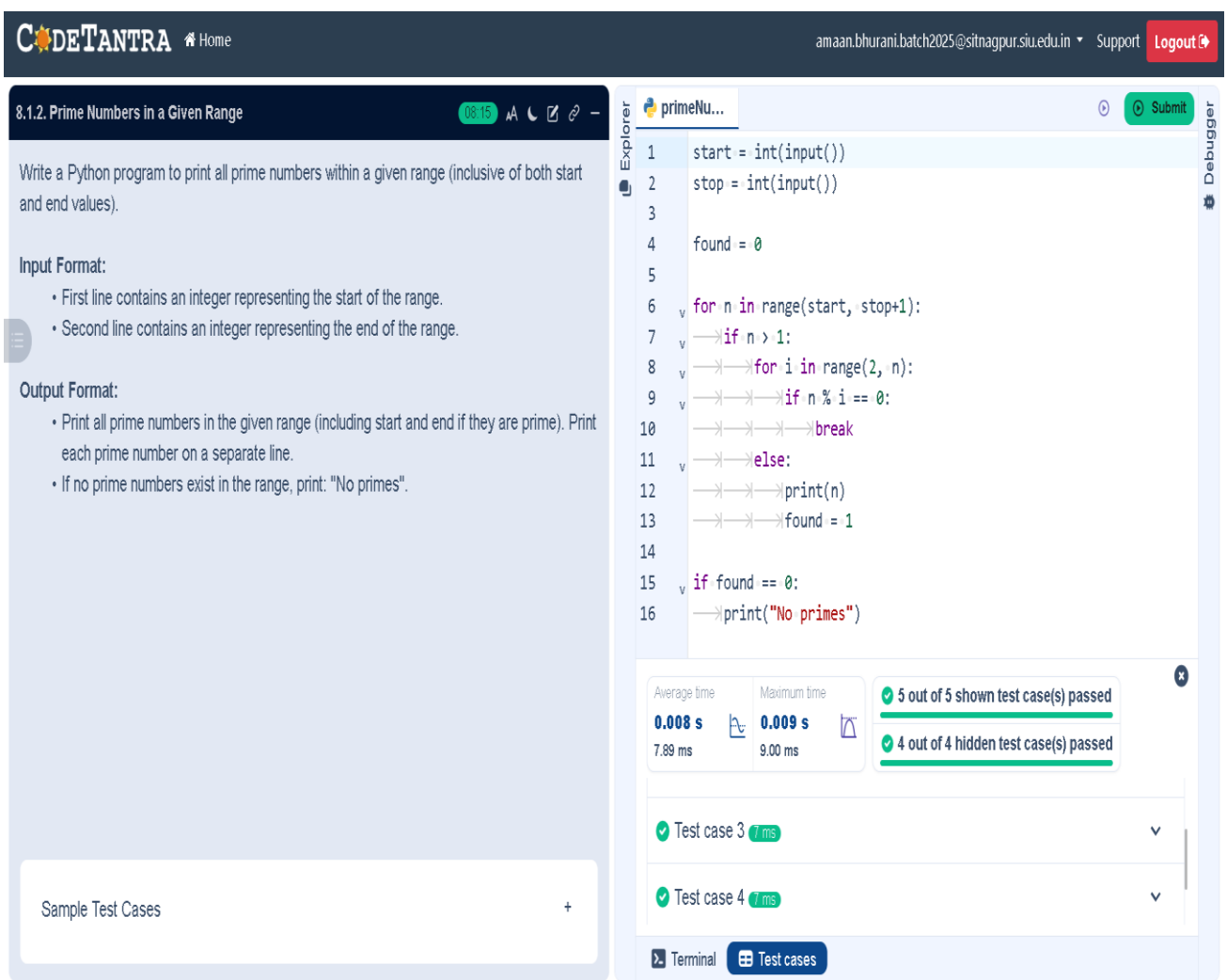
Flowchart :


Start
INPUT start , stop
Found = 0
n = start
n <= stop ?
False
True
n > 1 ?
False
True
i = 2
i < n ?
False
True
n % i == 0 ?
True
break
False
i = i + 1
found == 0 ?
True
Print n
found = 1
Print “No Primes”
n = n + 1
Stop

Code :

```
start = int(input())
stop = int(input())
found = 0

for n in range(start, stop+1):
    if n > 1:
        for i in range(2, n):
            if n % i == 0:
                break
        else:
            print(n)
            found = 1

if found == 0:
    print("No primes")
```

Execution :